JUNIOR – MID · FRONTEND DEVELOPER

Technical Assessment

SnippetVault — Public Code Snippet Manager

1–2 Days	Junior–Mid	Auth + Public
Estimated	Level	Features

What this assessment is about

We want to see how you think — how you structure components, handle auth flows, manage shared vs. private data, and deal with real UI edge cases. The project is scoped to be completable in 1 to 2 days. Quality and clarity matter more than completeness. Shipping something focused and clean beats a half-working feature dump.

01 Project Overview

You will build SnippetVault — a public code snippet manager. It is a tool developers use to save, tag, search, and share reusable bits of code. Think of it as a lightweight, focused alternative to GitHub Gists, with a clean public profile and rich sharing features.

The app has two distinct modes: an authenticated workspace where a user manages their own snippets, and a public-facing surface where anyone can browse, view, and copy shared snippets without logging in.

What you are building

- Email/password authentication with protected and public routes
- A personal dashboard to create, edit, search, and filter your snippets
- A public profile page showing all snippets a user has made public
- Four sharing mechanisms: public URL, share with specific users, image export, and clipboard link
- Syntax-highlighted code preview with copy-to-clipboard feedback

Scope note

Use Supabase Auth for authentication — do not build your own auth system. Row Level Security (RLS) policies on your tables are required to protect private snippets.

02 Core Requirements

2.1 Authentication

- Sign up page with email, password, and a display name field
- Log in page with email and password
- Form validation: all fields required, password minimum 8 characters, email format check
- On login success, redirect to /dashboard — on sign-up success, redirect to /dashboard with a welcome toast
- A persistent nav bar shows the logged-in user's name and avatar (initials fallback) with a logout option
- All /dashboard routes must redirect unauthenticated users to /login — use Next.js middleware for this
- Public routes (/s/[id], /u/[username]) must be accessible without login

2.2 Snippet Dashboard (authenticated)

- Display all of the current user's snippets as cards in a responsive grid
- Each card shows: title, language badge, first 3 lines of code, tags, and a visibility indicator (public / private)
- Clicking a card opens the detail side panel
- A '+ New Snippet' button in the header opens the create form
- Live search (debounced) and tag filter chips narrow the visible snippets client-side

2.3 Snippet Detail Panel

- Opens as a right-side Sheet panel without leaving the dashboard
- Full syntax-highlighted code view with a Copy button — shows 'Copied!' for 2 seconds then reverts, per snippet
- Displays title, language, description, tags, visibility status, and timestamps
- Edit and Delete actions — delete requires an inline confirmation step, not a browser dialog
- A Share button opens the share menu (see Section 2.5)

2.4 Create & Edit Form

- Fields: title (required), language (required, fixed select list), description (optional), code (required), tags (free-form chips), visibility toggle (Public / Private, default Private)
- Inline validation on submit — required fields show error messages below the input
- On success the list updates instantly without a full page reload
- Editing pre-fills all fields including the current visibility state

2.5 Sharing Features

All four sharing options must be accessible from the Share button in the detail panel. The share menu (a Popover or Sheet) contains:

a) Public URL

- Each snippet has a permanent public URL: /s/[snippet-id]

- The public page renders the snippet with full syntax highlighting, the author's display name, and a Copy Code button
- If the snippet is set to Private, the public URL must return a 404-style 'Snippet not found' page — never expose private code
- The share menu shows the public URL only if the snippet is currently set to Public; otherwise it shows a prompt to make it public first

b) Share with Specific Users

- An input field in the share menu accepts an email address
- Submitting adds that email to a snippet_shares table, granting read access
- Shared users can view the snippet at /s/[id] even if it is set to Private
- The share menu lists current shares with a Remove button next to each
- If the email does not match any registered user, show a clear error: 'No account found with that email'

c) Export as Image

- A button in the share menu generates a styled PNG of the code snippet
- Use html-to-image or dom-to-blob to capture a styled code block with the snippet title, language badge, and a SnippetVault watermark
- The image downloads automatically with the filename [snippet-title].png
- The export button must show a loading state while the image is being generated

d) Copy Shareable Link

- A single-click button copies the full public URL to clipboard
- Button label changes to 'Link Copied!' for 2 seconds then reverts
- If the snippet is Private and not shared, the copied link should still work for the owner — it must not work for anonymous visitors

2.6 Public Profile Page

- Route: /u/[username] — accessible without login
- Shows the user's display name, total public snippet count, and a grid of all their public snippets
- Each card links to the snippet's public page /s/[id]
- If the user has no public snippets, show an empty state: 'No public snippets yet'
- This page should be statically generated where possible (Next.js SSG or ISR)

03 Technical Constraints

All of the following are required. Do not substitute alternatives.

- **Next.js 16+** — App Router + Middleware for auth
- **TailwindCSS** — All styling
- **TanStack Query v5** — Server state & mutations
- **TypeScript** — Strict mode throughout

- **shadcn/ui** — Component library
- **Supabase** — Auth + Postgres + RLS policies
- **Zustand** — UI & filter state
- **html-to-image** — Image export (or dom-to-blob)
- **Syntax Highlighter** — Prism, Shiki, or react-syntax-highlighter

3.1 Architecture Expectations

- Next.js middleware must handle auth redirects — do not check auth inside every page component
- Supabase RLS policies are required — a user must never be able to read another user's private snippets via a direct API call, even if they guess the ID
- Custom hooks wrap all TanStack Query logic — no raw useQuery calls in JSX
- API layer in /lib/api.ts — no Supabase client calls scattered across components
- Types defined for all data shapes in /types/index.ts

3.2 UI Generation Requirement (Google Stitch)

To streamline the design process and evaluate how candidates work with modern AI-assisted design tools, the UI for this project must be generated using **Google Stitch**.

Requirement

Before starting development, candidates must generate the initial UI layouts using **Google Stitch** → **Figma**.

The following screens must be generated:

- Login Page
- Sign Up Page
- Dashboard (Snippet Grid)
- Snippet Detail Panel (Side Sheet)
- Create / Edit Snippet Form
- Share Menu
- Public Snippet Page `/s/[id]`
- Public Profile Page `/u/[username]`

The generated designs should include:

- Layout structure
- Component placement
- Responsive considerations (desktop / tablet / mobile)
- Code block area for syntax highlighting
- Snippet cards and metadata sections

Implementation

The generated UI should then be implemented in the project using:

- **shadcn/ui components**
- **TailwindCSS**

The implementation does **not need to be pixel-perfect**, but it should follow the general layout and structure of the generated design.

Design Submission

Candidates must share the generated **Google Stitch / Figma design file** with the following emails:

- avinash.patel@weframetech.com
- ayush.bathre@weframetech.com
- harsh.raj@weframetech.com

04 Database Schema

Create the following tables in Supabase. Enable Row Level Security on all tables.

profiles (extends Supabase auth.users)

Column	Type	Constraint	Notes
`id`	uuid	FK → auth.users	Matches Supabase auth user ID
`username`	text	UNIQUE NOT NULL	URL-safe display name for /u/[username]
`display_name`	text	NOT NULL	Full name shown in the UI
`avatar_url`	text	NULLABLE	Optional profile image URL
`created_at`	timestamptz	DEFAULT now()	Account creation timestamp

snippets

Column	Type	Constraint	Notes
`id`	uuid	PRIMARY KEY	Auto-generated
`user_id`	uuid	FK → profiles.id	Owner of the snippet
`title`	text	NOT NULL	Short descriptive title
`language`	text	NOT NULL	e.g. typescript, python, bash
`description`	text	NULLABLE	Optional plain-text note
`code`	text	NOT NULL	Raw code content
`is_public`	boolean	DEFAULT false	Controls public visibility
`created_at`	timestamptz	DEFAULT now()	Creation timestamp
`updated_at`	timestamptz	DEFAULT now()	Updated on every edit

tags + snippet_tags

Column	Type	Constraint	Notes
`id`	uuid	PRIMARY KEY	Auto-generated tag ID
`name`	text	UNIQUE NOT NULL	Lowercase tag label

Column	Type	Constraint	Notes
`snippet_id`	uuid	FK → snippets.id	References snippet
`tag_id`	uuid	FK → tags.id	References tag

snippet_shares

Column	Type	Constraint	Notes
`id`	uuid	PRIMARY KEY	Auto-generated
`snippet_id`	uuid	FK → snippets.id	Shared snippet
`shared_with`	uuid	FK → profiles.id	Recipient user
`created_at`	timestamptz	DEFAULT now()	Share timestamp

RLS Policy Requirements

snippets: SELECT allowed if is_public = true OR user_id = auth.uid() OR id exists in snippet_shares for the requesting user. INSERT/UPDATE/DELETE allowed only if user_id = auth.uid(). snippet_shares: SELECT/INSERT/DELETE allowed only for the snippet owner (user_id = auth.uid()).

Seed data

Include at least 8 seed snippets across different languages. Mix public and private. At least 2 should have snippet_shares entries so reviewers can test the shared-access flow.

05 State Management

Server state lives in TanStack Query. UI state lives in Zustand. These must not be mixed.

TanStack Query — Server State

- **useQuery(['snippets', userId]) —** fetch the current user's snippets with tags. staleTime: 60s.
- **useQuery(['snippet', id]) —** single snippet for the detail panel
- **useQuery(['shares', snippetId]) —** list of users a snippet is shared with
- **useMutation (create / update) —** on success, invalidate ['snippets', userId]
- **useMutation (delete) —** optimistic removal from list cache; roll back on error with a toast
- **useMutation (addShare / removeShare) —** invalidate ['shares', snippetId] on settle

Zustand — UI State

- **activeSnippetId: string | null —** drives which detail panel is open
- **isShareMenuOpen: boolean —** controls the share popover
- **searchQuery: string —** live search input value
- **selectedTags: string[] —** active tag filter chips
- Filtered list derived via useMemo from server data + store values — no re-fetch on filter change

06 UI & shadcn Requirements

Required shadcn Components

- Button, Input, Textarea, Select — all form controls
- Sheet — snippet detail side panel
- Dialog — delete confirmation
- Popover — share menu
- Badge — language label, tag chips, public/private indicator
- Skeleton — loading state for snippet grid
- Toaster / useToast — copy feedback, share success, and error messages
- Switch — the Public / Private visibility toggle in the create/edit form
- Separator — divides sections within the share menu

Responsive Layout

- Desktop (≥ 1024px): 3-column snippet grid, detail panel alongside
- Tablet (768–1023px): 2-column grid, detail panel overlays with backdrop
- Mobile (< 768px): single column, detail panel is full-screen
- Public profile and snippet pages must look good without login — treat them as marketing-facing

07 Edge Cases to Handle

Auth Edge Cases

- Expired session: user should be silently redirected to /login, not shown a broken page
- Signing up with an already-registered email: show 'An account with this email already exists'
- Accessing /dashboard while logged out: redirect to /login with a ?redirect= param so they return after login

Sharing Edge Cases

- Sharing with your own email: show 'You cannot share a snippet with yourself'
- Sharing with an email not in the system: show 'No account found with that email'

- Trying to export image on a very long snippet: cap the export at 50 lines and add a '...truncated' indicator
- Making a snippet private after it has been shared via URL: the public URL must immediately return 404 — but shared-with users retain access

General UI Edge Cases

- Copy button: works on both http and https; has a textarea fallback for older browsers
- Filter with no results: friendly empty state + 'Clear filters' button
- Failed delete: optimistic rollback + toast with retry option
- Failed image export: toast with the error message, button returns to default state

08 Evaluation Criteria

Area	What We Assess	Weight
Auth & Security	Correct Supabase Auth flow, RLS enforced, middleware redirects, no private data leaks	**25%**
Functionality	All core features work: CRUD, search/filter, all 4 share types, public pages	**25%**
State Handling	TanStack Query + Zustand used correctly, optimistic delete with rollback	**20%**
UI Polish	shadcn usage, responsive layout, all loading/empty/error states handled	**15%**
Code Quality	TypeScript strictness, clean hooks, isolated API layer, feature-based structure	**15%**

Security is evaluated seriously

We will test RLS directly by calling the Supabase API with a different user's auth token and attempting to read private snippets. If private data is accessible, this will be a significant mark-down regardless of how good the UI looks.

09 Bonus Challenges

Only attempt these after all core requirements are solid.

Bonus A — View Count

Track how many times a public snippet page (/s/[id]) has been viewed. Show the count on the detail panel and the public page. Increment must happen server-side (a Server Action or API route) — not from the client directly to avoid RLS bypass.

Bonus B — Keyboard Navigation

Arrow keys navigate between snippet cards. Enter opens the detail panel. Escape closes it. Tab moves focus into the panel. All interactions must work without a mouse.

Bonus C — Language Stats Bar

A small horizontal bar in the dashboard header shows snippet counts per language as proportional colored segments. Clicking a segment filters the list. Built entirely with Tailwind — no chart library.

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10 Submission

- Push your code to a **private GitHub repository**.
- Grant repository and supabase access to the following emails:
  - avinash.patel@weframetech.com
  - ayush.bathre@weframetech.com
  - harsh.raj@weframetech.com
- The **README must include**:
  - Setup steps
  - Required environment variables
  - Supabase project configuration notes
  - Any architectural decisions or known trade-offs
- **Deploy a live demo to Vercel** — this is required, not optional.
- **Never commit Supabase credentials**. Use `.env.local` for secrets and document the environment variable names in the README.

Time limit

Please submit by **March 11th, 7:00 PM IST**. Applications submitted after this time will be rejected. If you need more time for any reason, let us know **before the deadline**—it will not count against you.

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Good luck — keep it simple, keep it clean.